

Reconstructing Humans and Objects in Interaction using Large Reconstruction Models

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Abstract. Estimation of Human-Object Interactions in 3D (3D HOI) is a fundamental problem in 3D computer vision with broad applications in AR/VR, robotics, and embodied AI. However, HOI estimation remains challenging due to depth ambiguities, occlusions, and object shape variability. Existing approaches are primarily concerned with reprojection and contact constraints, fitting parametric human models and object templates to the image and/or satisfying contact constraints. In this paper, we explore a different avenue. We present MILO, a framework that leverages the visual capabilities of Large Reconstruction Models (LRM) to recover detailed 3D human-object reconstructions from a single image. Our key observation is that LRMs provide a powerful geometric scaffold that preserves relative human-object arrangement and proximity cues. This significantly simplifies the reconstruction procedure, reframing the problem as interpreting the LRM mesh: we segment it into human and object components, fit a parametric body model to the human part, and optionally align an object template to the object part (if such a template is available). MILO achieves strong reconstruction accuracy and outperforms existing baselines across multiple benchmarks and interaction scenarios. Our code will be released to facilitate future research.

Keywords: 3D · human-object interaction · Large Reconstruction Models

1 Introduction

Daily human life is characterized by rich 3D interactions with a wide variety of objects, from utensils and smartphones to furniture and tools. Modeling these human-object interactions (HOI) is central to many vision applications, including immersive AR/VR, teleoperation, and robot manipulation. However, HOI reconstruction from a single image is difficult due to diverse object shapes and affordances, multiple interaction modes, occlusions, and depth ambiguities, which make the recovery of human pose and object geometry particularly challenging.

Human parametric models such as SMPL [36], MANO [50], and SMPL-X [44] provide standardized representations which have led to successes for 3D reconstruction and of humans. In contrast, the absence of analogous canonical models for generic objects renders object reconstruction significantly more challenging. Moreover, the generalization capabilities of methods [42, 64] trained on existing HOI datasets [2, 19] are hindered by the characteristics of these datasets, which



Fig. 1: We present MILO, an approach for reconstructing 3D human-object interactions from a single image. Given an input photo (left), MILO uses a Large Reconstruction Model to predict a combined human-object mesh (center). This mesh is interpreted (right) by identifying human and object parts, fitting a SMPL-H body to the human part, and optionally an object template to the object part. The examples illustrate accurate 3D human pose and shape together with consistent object pose & shape. Human meshes are shown in blue and objects in red.

are captured in controlled environments with a limited set of objects and scripted interactions. Additionally, approaches [5, 65] that require the retrieval of object meshes further depend on the quality and coverage of available shape repositories. More importantly, accurate placement of retrieved meshes often requires privileged cues such as ground-truth contact or depth [5, 42, 71].

Our insight is that we can circumvent many of these limitations by leveraging the rich visual capabilities of Large Reconstruction Models (LRMs) [17, 21, 58, 68]. LRMs are trained on large-scale data [6] and are capable of reconstructing complex scenes, providing an image-specific joint reconstruction of the human and the object. This effectively provides a human-object geometric scaffold encoding relative arrangement and proximity cues. Building on this, our method replaces ambiguous image fitting and contact assumptions with the interpretation of an LRM-generated human-object reconstruction.

Our goal is to systematically assess the extent to which LRMs can be used for HOI reconstruction. While previous work has used LRMs primarily to reconstruct the object [35], our insight is that LRMs can be very powerful for capturing the relative positioning and proximity cues between a human and an object. To this end, we propose MILO; Modeling Interactions using Large Reconstruction MOdels. MILO first applies an off-the-shelf LRM [58] to the input RGB image to obtain a holistic human-object mesh. This reframes our problem as explaining the LRM interaction mesh. Through multi-view segmentation of the mesh, we can extract the human and the object point cloud. For the human part, we fit the parametric SMPL-H body model to the LRM mesh via predicted 3D keypoints [45, 69]. For the object, we compose the raw LRM reconstruction with the fitted human, or optionally align a template mesh if one is available. In that case, we establish semantic correspondences [72], which allow us to align the template to the LRM mesh. We present example results in Fig. 1.

We quantitatively compare MILO with state-of-the-art methods on multiple datasets, i.e., InterCap [19], HODome [73] and IMHD [75], while also showing qualitative results on in-the-wild images [5]. Importantly, MILO outperforms previous work across the board, without using ground-truth contact information.

Our main contributions are as follows:

1. We demonstrate the power of Large Reconstruction Models for HOI reconstruction. Instead of operating with the high ambiguity of reprojection objectives, we reframe the problem as explaining the LRM mesh results.
2. We design an approach that robustly fits a parametric human model to the LRM mesh and optionally aligns an object template to it.
3. We achieve state-of-the-art performance across multiple benchmarks, all while using weaker information than previous work, i.e., without relying on contact information.

2 Related Works

2.1 Monocular 3D Human Reconstruction

Single-image 3D human reconstruction methods can be broadly categorized into parametric and non-parametric approaches. Parametric methods [8, 25, 28, 29, 51] estimate the parameters of a body model [36, 44, 50, 67], whereas non-parametric methods [26, 31, 32] predict explicit meshes or point clouds. Optimization-based pipelines [3, 27, 44, 67] recover body pose and shape by fitting a parametric body model to 2D cues, e.g., keypoints, silhouettes, or segmentation, via differentiable rendering. In contrast, learning-based approaches directly regress body-model parameters from inputs [9, 22, 29, 30, 40] using deep networks. While these methods have led to robust monocular human reconstruction, they typically focus on the person in isolation and do not explicitly reason about surrounding objects or human-object contact. In our method, we use the HMR2.0 network [14] to obtain initial SMPL parameters from the input RGB image.

2.2 Monocular 3D Object Reconstruction

Monocular 3D object reconstruction has a long history with a wide range of 3D representations explored. Early work employed explicit volumetric [4, 13], point-based [10], or mesh-based [10, 62] representations, while later methods [39, 43] adopted implicit signed-distance or radiance-field representations. More recently, 3D Gaussian splatting [24] has emerged as an alternative to NeRF-style radiance fields, enabling efficient object reconstruction. Diffusion-based image-to-3D approaches [38, 47, 56] and view-conditioned diffusion models [33] further improve visual fidelity and handle complex appearance, but often rely on per-instance optimization or suffer from multi-view inconsistencies across generated views.

Large Reconstruction Models [17, 21, 55, 59, 63] combine 3D representations, such as triplanes with multi-view image supervision, enabling efficient and view-consistent reconstruction of diverse categories. MILO adopts Hunyuan3D-2.0 [58],

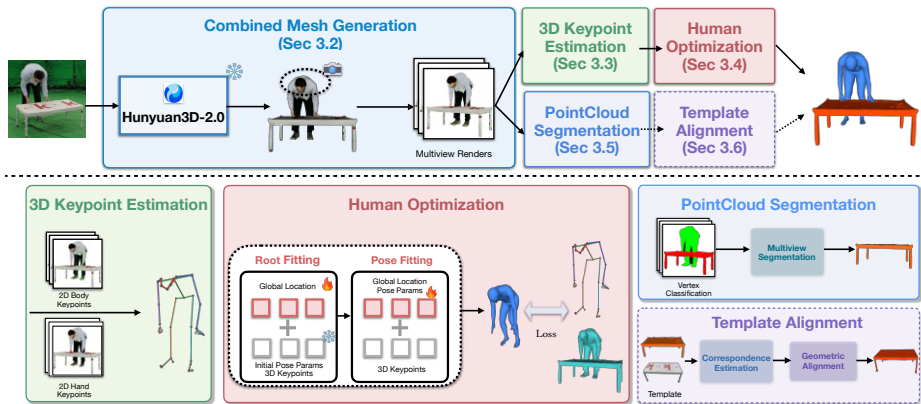


Fig. 2: Overview of MILO, our method for reconstructing 3D human-object interactions from a single image. Given an input RGB image, we first use Hunyuan3D-2.0 [58] to predict a combined human-object mesh (Sec. 3.2). We then render the mesh from multiple virtual views and estimate 2D body and hand keypoints in each view (Sec. 3.3), which are robustly triangulated to obtain 3D keypoints on the LRM reconstruction. Starting from an initial SMPL-H estimate from HMR2.0 [14] and HaMeR [45], we optimize SMPL-H in two stages (Sec. 3.4): (i) *root fitting*, optimizing only global orientation and translation, and (ii) *pose fitting*, additionally optimizing body/hand pose, and shape, regularized by pose/shape priors and a visibility-aware 3D consistency term. Finally, we perform multi-view segmentation of the LRM mesh to extract the object component and compose it with the optimized human mesh for evaluation and visualization (Sec. 3.5). *If an object template is available*, we optionally align it to the segmented object using semantic correspondences (Sec. 3.6).

which has a DiT component that predicts a base 3D mesh, and a diffusion model that conditions on geometry cues to synthesize multi-view images for texture baking. By leveraging the LRM, we obtain an image-specific joint human-object mesh, which captures accurate geometry and proximity information.

2.3 Monocular 3D Human-Object Reconstruction

Joint reconstruction of humans and objects from a single image has been explored with both learning- and optimization-based approaches. HOI-TG [64] employs a transformer to jointly reason about pose and interaction. LEMON [70] learns 3D human-object interaction relations from 2D images by jointly predicting human contact, object affordance, and spatial relations. CONTHO [42] predicts vertex-level contact maps and refines the human-object mesh poses accordingly. StackFLOW [20] regresses dense human-object offsets, and refines both the human and object reconstructions using stacked normalizing flows. CHORE [66] jointly learns a part-correspondence field that links human and object geometry. TriDi [46] supports multiple conditioning modes by introducing a trilateral diffusion model over humans, objects, and interactions.

A common strategy involves reconstructing the object by retrieving the closest object mesh from existing shape repositories [7]. PICO [5] retrieves object

candidates, Wu et al. [65] uses text-to-3D retrieval to obtain an object template. More related to our work, LRMs can also be leveraged to reconstruct objects of interaction directly from images, without relying on explicit CAD templates. For instance, EasyHOI [35] employs InstantMesh [68] to obtain object meshes in hand-object interaction scenarios. Our approach is related to the use of LRMs, but we propose to use these powerful models for joint human-object reconstruction, which makes the relative positioning of humans and objects easier.

Most of the previous methods assume access to object templates and/or GT contact annotations, which constrains scalability and generalization to novel objects and in-the-wild scenes. Our approach makes the use of an object template optional and does not rely on explicit contact information.

2.4 Human-Object Contact Estimation

Human-object contact provides a powerful geometric cue that tightly constrains the arrangement of humans and objects. Early work modeled contact at a coarse level [1,16,23], over patches [11,41], or only on the human body surface [49,53,74]. More recent approaches [15,18,54,60,71] estimate dense, vertex-level contact or correspondence maps and exploit them to regularize HOI reconstruction or scene understanding tasks. Several HOI methods integrate contact signals directly into their pipelines. CONTHO [42] uses vertex-level human-object contact maps for mesh refinement. CHORE [66] learns part-to-body correspondence fields that implicitly encode contact patterns. PICO [5] uses paired human-object contact maps to guide joint human-object fitting. A related effort, HULC [52], exploits body-scene contact to stabilize monocular motion capture. In contrast to them, our method does not require contact annotations. Instead, we show that we can extract competitive contact estimates from our HOI reconstructions.

3 Technical Approach

In this section, we present MILO, an approach that reconstructs detailed human-object interactions from a single RGB image by leveraging an LRM. Sections 3.1 and 3.2 discuss preliminaries, while Sections 3.3 and 3.4 describe how we can fit a parametric human model to the LRM mesh. In Section 3.5, we describe how we can segment the reconstructed mesh to a human and object part. Finally, if an object template is available, Section 3.6 describes how we can align it to the LRM reconstruction. An overview of our pipeline is shown in Fig. 2.

3.1 Preliminaries

We represent the human body with the SMPL-H model [50], parameterized by global orientation $\Phi \in \mathbb{R}^3$, body pose $\Theta \in \mathbb{R}^{22 \times 3}$, hand pose $\Theta_h \in \mathbb{R}^{15 \times 3}$, shape $\beta \in \mathbb{R}^{10}$, and root translation $T \in \mathbb{R}^3$. Given these parameters, SMPL-H outputs body vertices and kinematic joints as: $[V, J] = \mathcal{M}(\Phi, \Theta, \Theta_h, \beta, T)$ where $V \in \mathbb{R}^{3 \times 6890}$ are the mesh vertices and $J \in \mathbb{R}^{3 \times 52}$ are the joint locations.

3.2 Human-Object Interaction

One of our key ideas is to use an LRM to directly reconstruct a combined human-object mesh from the input image. Concretely, we employ Hunyuan3D-2.0 [58], which produces a combined mesh capturing both the person and the interacted object. This choice is driven by two key advantages: (1) LRMs can generate a high-quality 3D approximation of the object and interaction present in the image. (2) The combined mesh encodes rich information about human-object spatial relations and proximity cues without requiring explicit contact annotations. To this end, we pass an RGBA image to the LRM to obtain a holistic human-object mesh. The alpha channel represents the combined segmentation maps of the human and the object. We interpret the LRM output mesh as a non-parametric interaction scaffold, segment it into human and object components, and fit the SMPL-H model to the human part to explain how the person interacts with the object. If an object template is available, we can also align it to the LRM mesh.

3.3 3D Keypoint Estimation

The goal of this step is to estimate a sparse set of 3D keypoints on the LRM mesh and use them as observations for subsequent SMPL-H fitting. Specifically, we render the LRM mesh from a set of virtual views $\mathcal{V}$ (set to 60) with known camera parameters and run ViTPose [69] and HaMeR [45] on each rendered image to obtain body and hand keypoints, respectively. For each view $v \in \mathcal{V}$, ViTPose predicts OpenPose COCO-25 body keypoints $x_b^{(v)} \in \mathbb{R}^{25 \times 2}$ with confidence scores $c_b^{(v)} \in [0, 1]^{25}$. Simultaneously, HaMeR predicts hand keypoints $x_{h,s}^{(v)} \in \mathbb{R}^{21 \times 2}$ for $s \in \{L, R\}$ (we assume confidence of 1).

Confidence filtering. For each keypoint index k , we keep only views whose confidence exceeds a threshold γ (empirically set to 0.6): $\mathcal{O}_k = \left\{ v \in \mathcal{V} \mid c_k^{(v)} > \gamma \right\}$.

If $|\mathcal{O}_k| < N_{\min} = 3$, the keypoint is marked as invalid and excluded from fitting.

Robust triangulation. For each keypoint, we triangulate candidate 3D points using all pairs of observed views. We select the hypothesis with the largest consensus set, i.e., the most inliers under reprojection-error threshold τ (set to 5.0). The final 3D point is obtained by nonlinear refinement over the inlier views ($\mathcal{I}_k^*$):

$$X_k^* = \arg \min_{X \in \mathbb{R}^3} \sum_{v \in \mathcal{I}_k^*} \left\| \pi_v(X) - x_k^{(v)} \right\|_2^2. \quad (1)$$

Confidence aggregation and output. For each valid keypoint, we compute an aggregated reliability score as the mean confidence over its inlier views: $\bar{c}_k = \frac{1}{|\mathcal{I}_k^*|} \sum_{v \in \mathcal{I}_k^*} c_k^{(v)}$. Applying this to body and hand joints yields $X_b \in \mathbb{R}^{25 \times 3}$, $X_{h,L} \in \mathbb{R}^{21 \times 3}$, and $X_{h,R} \in \mathbb{R}^{21 \times 3}$, which we concatenate into $X = [X_b; X_{h,L}; X_{h,R}] \in \mathbb{R}^{67 \times 3}$ and $\bar{c} \in \mathbb{R}^{67}$. The pair $(X, \bar{c})$ defines a sparse, confidence-weighted set of 3D keypoints that are used in the next stage to fit SMPL-H to the LRM mesh.

3.4 Human Optimization

We initialize SMPL-H parameters from the input image using HMR2.0 [14], which predicts human pose and shape in the camera frame. We also initialize the hand pose parameters using HaMeR [45]. We then refine this estimate in two stages: the root fitting, which optimizes for the global orientation and the root translation, followed by the pose fitting stage, which additionally optimizes for the body shape, body and hand pose. Both stages optimize the SMPL-H mesh, keeping the LRM mesh fixed. Additionally, in the first fitting stage, we optimize a human scale parameter. This helps align the human to the LRM mesh without devolving to degenerate scales, since the LRM mesh is not guaranteed to produce the combined mesh at the correct scale.

Root Fitting. Let $\hat{P} = \{\tilde{\Phi}, \tilde{\Theta}, \tilde{\beta}, \tilde{\Gamma}\}$ denote the initial SMPL-H prediction for image I_i as given by HMR2.0 [14], with the parameters expressed in the camera frame. We initialize the body pose (Θ), hand pose (Θ^h) and body shape (β) with these values. In this first stage, we keep Θ, Θ^h, β fixed and only optimize Φ and Γ to align the SMPL-H joints X^h to their corresponding 3D keypoints X estimated in the previous section. The objective is expressed as: $\mathcal{L}_{rf} = \sum_i \rho(X_i^h - X_i)$, where ρ is the Geman-McClure robustifier [12]. This stage runs for 30 iterations and coarsely aligns the SMPL-H mesh with the LRM mesh.

Pose Fitting. Next, we optimize the full SMPL-H parameter set, i.e., shape β_i , body pose Θ_i , and hand pose Θ_i^h . To keep the solution feasible, we regularize pose and shape with the VPoser prior [44] and an L_2 shape prior respectively. Let $\zeta_{bp} \in \mathbb{R}^{32}$ be the representation of the body pose parameters Θ_i in the latent space of VPoser [44], and $\zeta_{hp} \in \mathbb{R}^{15}$ be the representation of the hand pose parameters Θ_i^h in the latent space of the MANO model [50]. The regularization terms can be expressed as: $\mathcal{L}_{bp} = \|\zeta_{bp}\|^2$, $\mathcal{L}_{hp} = \|\zeta_{hp}\|^2$ and $\mathcal{L}_\beta = \|\beta\|^2$.

Under severe self-occlusion or object occlusion, the LRM may hallucinate missing body parts, which can lead to image-inconsistent pose estimates. While $\mathcal{L}_{bp}$ and $\mathcal{L}_{hp}$ encourage feasible poses, they do not explicitly preserve consistency with the image. To improve robustness in such cases, we add an anchoring loss based on the HMR2.0 [14] pose initialization. We use the initialized latent body pose ζ_{bp}^{hmr} and penalize large deviations from it during pose fitting:

$$\mathcal{L}_{ha} = \sum_j \left[(\zeta_{bp,j} - \zeta_{bp,j}^{\text{hmr}})^2 - \tau \right]_+, \quad (2)$$

where τ is a tolerance margin, set to 2.5. This acts as a soft anchor: small latent refinements are not penalized, while larger deviations are discouraged.

To further improve alignment in occluded regions, we introduce a visible-vertex 3D consistency loss with the LRM reconstruction. Let $P^{\text{obs}} \subset \mathbb{R}^3$ denote a set of 3D points sampled from the observed (human) LRM reconstruction, and let $P^{\text{pred}} \subset \mathbb{R}^3$ denote the set of predicted SMPL-H mesh vertices. Because the

reconstruction can be unreliable in occluded areas, we restrict the comparison to a subset $P_{\mathcal{V}}^{\text{pred}} \subseteq P^{\text{pred}}$ containing only those predicted vertices associated with visible joints, where $\mathcal{V}$ is the vertex-visibility set estimated from the rendered views. We then define a one-way robust Chamfer loss from the observed points to these visible predicted vertices:

$$\mathcal{L}_{3D} = \frac{1}{|P^{\text{obs}}|} \sum_{p \in P^{\text{obs}}} \rho \left(\min_{q \in P_{\mathcal{V}}^{\text{pred}}} \|p - q\|_2 \right), \quad (3)$$

where $p \in P^{\text{obs}}$ is an observed 3D point, $q \in P_{\mathcal{V}}^{\text{pred}}$ is a visible predicted vertex, $\|\cdot\|_2$ denotes the Euclidean distance, $|P^{\text{obs}}|$ is the number of observed points, and $\rho(\cdot)$ is a robust penalty (Geman–McClure) to reduce sensitivity to outliers. This term improves fitting in both self-occlusion and object-occlusion cases.

We combine these with the root loss to obtain the final fitting objective:

$$\mathcal{L}_{pf} = \lambda_{rf} \mathcal{L}_{rf} + \lambda_{bp} \mathcal{L}_{bp} + \lambda_{hp} \mathcal{L}_{hp} + \lambda_{\beta} \mathcal{L}_{\beta} + \lambda_{ha} \mathcal{L}_{ha} + \lambda_{3D} \mathcal{L}_{3D}, \quad (4)$$

where $\lambda_{rf} = 10.0$, $\lambda_{bp} = \lambda_{hp} = 0.04$, $\lambda_{\beta} = 0.05$, $\lambda_{ha} = 10.0$ and $\lambda_{3D} = 50.0$. We optimize this objective for 60 iterations, yielding a human mesh that is tightly aligned with the LRM reconstruction while remaining anatomically plausible.

3.5 Point Cloud Segmentation

The next step is to extract the object geometry from the LRM mesh to obtain an object point cloud, which is composed with the SMPL-H human. We use the same multi-view renders from Sec. 3.3 and apply a segmentation model [34, 48] to obtain human and object masks. The prompt used is "person. <obj_name>". Visible vertices in each view are projected onto the image plane and labeled using these masks, followed by multi-view aggregation and filtering.

For each viewpoint k , we first compute a viewpoint quality score Q_k that reflects both geometric coverage and segmentation reliability:

$$Q_k = \alpha \cdot \frac{|V_k^{\text{visible}}|}{|V|} + (1 - \alpha) \cdot \frac{|M_k^{\text{human}}| + |M_k^{\text{object}}|}{|I_k|}, \quad (5)$$

where V_k^{visible} are the vertices visible in view k , V is the full vertex set, M_k^{human} and M_k^{object} are the human and object masks respectively, I_k is the set of image pixels, and $\alpha \in [0, 1]$ balances visibility and mask quality. The viewpoint quality takes into account both the ratio of vertices visible in that viewpoint and the quality of the human and object masks.

To better handle contact regions, we construct a multi-scale human-object boundary by dilating both masks with kernel sizes of 3, 5, and 7 and intersecting the dilated regions. We compute a boundary closeness weight using Euclidean distance transform from the boundary, normalized by a maximum distance threshold, and transformed through a non-linear function with an optional base score boost for highly overlapped regions. This penalizes vertices

near the boundary more heavily, while still allowing those farther away to be segmented. The optional boost is applied when the size of the object is much smaller compared to the human - the boundary region covers most of the object. For a projected vertex p_i , with distance $d(p_i, B)$ to this boundary B , we define a boundary-closeness weight:

$$w_i^{\text{boundary}} = \left(\min \left(\frac{d(p_i, B)}{d_{\max}}, 1 \right) \right)^\gamma + b, \quad (6)$$

where $d_{\max}$ is a distance threshold for normalization, $\gamma \in [0.3, 0.8]$ is an adaptive exponent determined by the human-object overlap ratio, and $b \geq 0$ is an optional base boost for highly overlapped regions. This encourages accurate labeling near the human-object contact regions. Finally, for each vertex V_i , we aggregate scores over all viewpoints K_i where it is visible:

$$w_i = \sum_{k \in K_i} Q_k \cdot w_i^{\text{boundary}}(V_i), \quad (7)$$

and apply a threshold on $\{w_i\}$ to obtain final human and object labels. Vertices labeled as non-human form the raw object point cloud. We then apply geometric filtering: statistical outlier removal, pruning vertices with too few neighbors within a local radius, and keeping only the largest DBSCAN cluster (more details are included in the Sup. Mat.). The resulting clean object point cloud is composed with the fitted SMPL-H mesh.

3.6 Template Alignment

MILO can operate without a template at test time. When a dataset provides a template mesh, we can optionally align it to the segmented object component of the LRM reconstruction. Let $M^{\text{tmp}} = \{u_i\}_{i=1}^N$ be the template mesh vertices and let $P^{\text{lrn}} = \{v_j\}_{j=1}^M$ denote the segmented LRM point cloud. We seek a similarity transform: $T(u) = s\mathbf{R}u + \mathbf{t}$, with $s > 0$, $\mathbf{R} \in SO(3)$, and $\mathbf{t} \in \mathbb{R}^3$, aligning the template to the reconstructed object. *Semantic correspondences.* Our first step is to establish correspondences between the LRM and the template mesh. We do this by considering multi-view renders of both meshes. For this, we first restrict matching to LRM renders that are most consistent with the input image. Using the geometry-aware semantic correspondence model of Zhang et al. [72], we compute a global mutual-nearest-neighbor similarity $S(\cdot, \cdot)$ between the input image and every LRM render, (considering only the object foreground), and retain the top- K_{in} views with $K_{\text{in}} = 5$.

Next, for each template render I_q^{tmp} , we evaluate its similarity $S(I_q^{\text{tmp}}, I_r^{\text{lrn}})$ against the retained LRM renders I_r^{lrn} and keep the top- K_{pair} matches ($K_{\text{pair}} =$

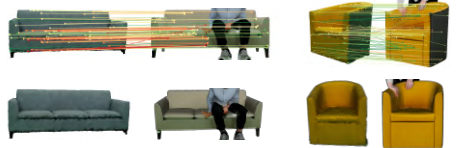


Fig. 3: Template Alignment. Top row: Visualization of semantic correspondences between a render of the template mesh and the LRM mesh. Bottom row: Alignment of the template mesh with the LRM mesh.

3). For each selected pair (q, r) , we compute dense bidirectional pixel correspondences in descriptor space. Let $\Pi_{q \rightarrow r}(x)$ denote the target pixel matched to source pixel x , and let $C_{q \rightarrow r}(x)$ be its cosine similarity.

We lift these 2D matches to 3D correspondences by unprojection. A template vertex u_i is considered only if it is visible in view q , projects inside the foreground mask, and its matched pixel $\Pi_{q \rightarrow r}(\pi_q(u_i))$ lies within a radius τ_{2D} of some visible LRM vertex projection in view r . The matched LRM vertex is:

$$j^*(i, q, r) = \arg \min_{j \in \mathcal{V}_r^{\text{lrn}}} \|\pi_r(v_j) - \Pi_{q \rightarrow r}(\pi_q(u_i))\|_2, \quad (8)$$

subject to: $C_{q \rightarrow r}(\pi_q(u_i)) > \tau_{\text{loc}}$ and $\|\pi_r(v_{j^*}) - \Pi_{q \rightarrow r}(\pi_q(u_i))\|_2 < \tau_{2D}$. We use $\tau_{\text{loc}} = 0.6$ and $\tau_{2D} = 15$ pixels. A template vertex may obtain multiple candidate matches across view pairs. We retain only the highest-scoring one, with score:

$$\omega_{i,q,r} = C_{q \rightarrow r}(\pi_q(u_i)) \cdot S(I_q^{\text{tmp}}, I_r^{\text{lrn}}). \quad (9)$$

This gives a sparse forward map $i \mapsto j_i$ with weight $w_i = \max_{q,r} \omega_{i,q,r}$. To suppress unstable matches, we also compute the reverse map $\text{LRM} \rightarrow \text{template}$ and apply neighborhood-based cycle consistency. A forward match $u_i \mapsto v_{j_i}$ is retained only if the reverse map sends v_{j_i} back to a template vertex within a local neighborhood of u_i , defined relative to the local template mesh scale. This yields the final sparse correspondence set: $\mathcal{C} = \{(i_k, j_k, w_k)\}_{k=1}^K$, (Fig. 3).

Template alignment. Given sparse weighted correspondences $\mathcal{C} = \{(i_k, j_k, w_k)\}_{k=1}^K$, we first estimate an initial similarity transform $T(u) = s\mathbf{R}u + \mathbf{t}$ by solving a weighted Sim(3) Procrustes problem with a weighted Kabsch fit. We then refine T with an iterative weighted Sim(3) re-estimation that jointly uses $\mathcal{C}$ and nearest-neighbor constraints to the segmented LRM object point cloud P^{lrn} .

At iteration t , we transform template vertices as $\tilde{u}_i^{(t)} = T^{(t)}(u_i)$, assign each $\tilde{u}_i^{(t)}$ its closest point $p_i^{(t)} = \text{NN}(\tilde{u}_i^{(t)}, P^{\text{lrn}})$, and form an inlier set $\mathcal{V}_t = \{i \mid \|\tilde{u}_i^{(t)} - p_i^{(t)}\|_2 < \tau_t\}$ using a threshold τ_t that is progressively tightened across iterations. We then compute $T^{(t+1)}$ by a *single* weighted Kabsch-with-scale fit on the combined pairs $\{(u_{i_k}, v_{j_k})\}_{k=1}^K \cup \{(u_i, p_i^{(t)})\}_{i \in \mathcal{V}_t}$, using weights w_k for semantic pairs and a constant weight of 10 for each nearest-neighbor pair. This procedure retains the semantic alignment captured by $\mathcal{C}$ while improving surface-level agreement with the reconstructed object geometry.

4 Experiments

This section presents our empirical evaluation. Section 4.1 compares MILO with the state-of-the-art on the InterCap [19], HODome [73] and IMHD [75] datasets. We evaluate settings with and without access to a template. Sec. 4.2 evaluates the effect of MILO on the downstream task of estimating vertex-level contact on the human body. Section 4.3 analyzes and ablates the key design choices. Finally, Sec. 4.4 presents qualitative in-the-wild reconstructions on PICO-db [5].

Methods	Type	Needs	Needs	PA-CD _h	PA-CD _o	PA-CD _{h+o}
		Template?	Contact?	(cm) ↓	(cm) ↓	(cm) ↓
CONTHO [42]	Reg.	✓	✓	8.36	24.30	13.14
HOI-TG [64]	Reg.	✓	✗	8.22	25.05	14.63
PHOSA [71]	Opt.	✓	✓	10.07	23.36	13.38
PICO [5]	Opt.	✓	✓	7.43	21.85	10.33
Ours (w/ template)	Opt.	✓	✗	6.96	18.97	7.45
Ours (w/o template)	Opt.	✗	✗	6.85	20.74	9.36

Table 1: Quantitative comparison with the state of the art on the InterCap dataset [19]. We report Procrustes-Aligned Chamfer Distance (PA-CD) for the human mesh (PA-CD_h), object mesh (PA-CD_o), and the combined human-object pair (PA-CD_{h+o}); lower is better. Regression (Reg.) and optimization (Opt.) baselines are evaluated under the training protocol from Cseke et al. [5]. Our method, MILO, uses no contact information, yet achieves the best overall reconstruction accuracy.

Methods	PA-CD _h	PA-CD _o	PA-CD _{h+o}
	(cm) ↓	(cm) ↓	(cm) ↓
PICO [5]	13.12	14.44	10.07
Ours (w/ template)	9.50	12.97	6.68
Ours (w/o template)	<u>9.71</u>	<u>13.75</u>	6.38

Methods	PA-CD _h	PA-CD _o	PA-CD _{h+o}
	(cm) ↓	(cm) ↓	(cm) ↓
PICO [5]	15.81	17.70	13.24
Ours (w/ template)	<u>11.76</u>	<u>13.39</u>	<u>10.10</u>
Ours (w/o template)	8.99	9.71	6.98

Table 2: Quantitative evaluation on the HODOME dataset [73].

Table 3: Quantitative evaluation on the IMHD dataset [75].

4.1 Comparison with the state-of-the-art

InterCap [19] is an indoor multi-view RGB-D dataset with ground-truth human and object meshes, comprising ten human subjects and ten everyday objects in various interaction scenarios. We evaluate MILO for HOI reconstruction in this controlled setting and compare it with both regression-based [42, 64] and optimization-based methods [5, 71] following the protocol of Cseke et al. [5]. For the learning-based baselines, we use checkpoints trained only on BEHAVE [2], i.e., enforcing an out-of-domain evaluation. We report Procrustes-Aligned Chamfer Distance (PA-CD) for the human mesh, the object mesh, and the combined human-object pair. The rigid alignment is performed on the combined mesh; after alignment, PA-CD is computed separately for the human and object components as well as for the joint reconstruction. Quantitative results are presented in Tab. 1. MILO outperforms all previous work (even without access to a template), while relying on less privileged information than competing methods.

Additionally, we evaluate MILO on HODome [73] and IMHD [75] datasets, comparing against the best-performing baseline PICO. HODome and IMHD are also indoor multi-view RGB-D datasets with ground-truth human and object meshes. HODome consists of 10 human subjects and 20 object classes, while IMHD consists of 8 human subjects and 10 object categories with different interactions with each object. As can be seen from Tab. 2 and Tab. 3, MILO performs better than PICO across the board, in both modes.

The improvements from MILO are also evident in the qualitative results (Fig. 4), where MILO achieves better 3D alignment between the human and the object, with more accurate contact and fewer penetrations than prior approaches.

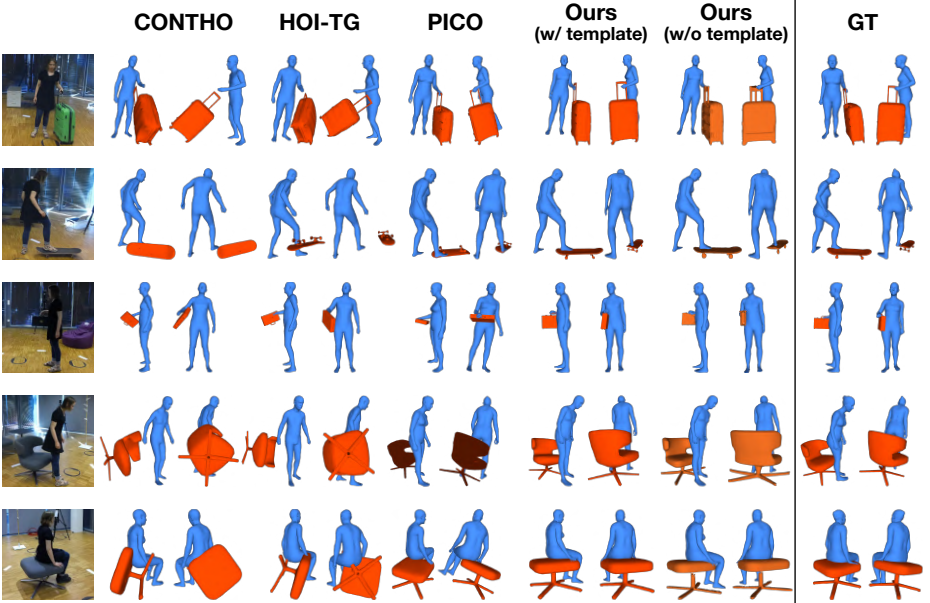


Fig. 4: Qualitative comparison on the InterCap dataset [19]. From left to right: input RGB image, CONTHO [42], HOI-TG [64], PICO [5], MILO reconstructions (with and without template meshes respectively), and ground-truth human-object meshes. Our method achieves better human-object alignment in 3D. Humans are rendered in blue, template objects in red and segmented object meshes in orange.

Method	F1↑	Precision↑	Recall↑	geo. (cm)↓
DECO [60]	9.37	12.57	10.28	126.70
Ours	30.00	36.57	39.49	39.57

Table 4: Contact evaluation. Metrics are reported on InterCap. Our 3D geometry-inferred contact outperforms the image-regressed contact by the DECO [60] baseline.

4.2 Contact Estimation

Although MILO does not predict contact explicitly, we can estimate the contact between the SMPL-H mesh and the aligned object template. Following this, we report the results in Tab. 4. Comparing with the DECO RICH checkpoint, MILO achieves higher F1, Precision and Recall, and lower geometric error on InterCap.

4.3 Ablation Experiments

In this section, we present multiple ablation studies. First, we evaluate the significance of the two fitting stages of the optimization pipeline. We perform this ablation using both the segmented LRM mesh and the aligned template (Tab. 5). As can be seen, removing the pose fitting, or both fitting stages, consistently degrades performance, confirming that both stages of our optimization are necessary to achieve the best overall reconstruction quality.

Then, we consider the use of alternative methods for point cloud segmentation. Specifically, in Table 6, we compare our proposed segmentation module with the recent P3-SAM method [37]. As can be seen, our strategy performs substantially better which justifies the design of our approach.

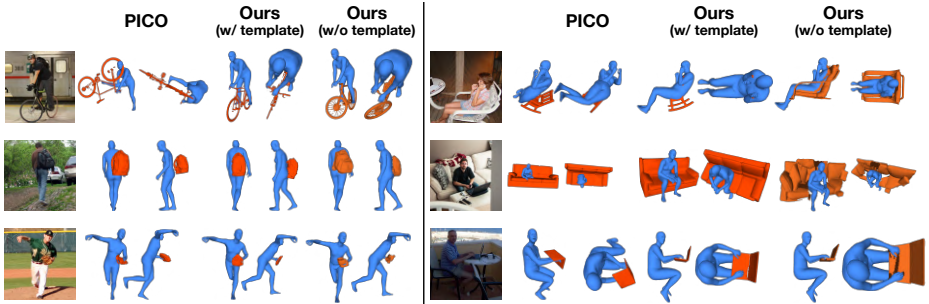


Fig. 5: Qualitative reconstructions on in-the-wild images from PICO-db [5]. MILO recovers 3D human-object geometry from a single image, with coherent arrangement and fewer interpenetrations than the PICO-fit [5] baseline. Humans are rendered in **blue**, template objects in **red**, and the segmented object mesh in **orange**.

Human Fitting	Uses Template	PA-CD _h (cm) ↓	PA-CD _o (cm) ↓	PA-CD _{h+o} (cm) ↓	Human Fitting	Uses Template	PA-CD _h (cm) ↓	PA-CD _o (cm) ↓	PA-CD _{h+o} (cm) ↓
No fitting	✗	30.58	23.53	14.10	No fitting	✓	14.97	37.02	14.57
Root Fitting	✗	7.98	22.53	10.15	Root Fitting	✓	7.79	18.63	7.99
Pose Fitting	✗	6.85	20.74	9.36	Pose Fitting	✓	6.96	18.97	7.45

Table 5: Effect of human fitting. Results are presented on the InterCap dataset [19]. **No fitting** computes the CD metrics using the raw LRM mesh (adopting our point-cloud segmentation), without SMPL-H optimization. **Root fitting** optimizes only the global orientation and translation of the human. **Pose fitting** additionally optimizes body/hand pose and shape for SMPL-H. We consider both template and template-free settings. The progressive reductions across metrics highlight the importance of the fitting stages, and the necessity of a parametric model, for precise 3D HOI reconstruction.

Next, in Tab. 7 we ablate the stages of our template alignment pipeline (Sec. 3.6). Using ICP on top of the initial weighted Kabsch fit consistently improves alignment across all metrics. The geometry-aware correspondences provide a reliable coarse similarity transform, while ICP further refines the local geometric fit to the segmented LRM object point cloud.

Finally, we ablate the point cloud segmentation pipeline introduced in Sec. 3.5. Specifically, we analyze the effect of (i) visibility coverage and mask quality, which together define the viewpoint quality score Q , and (ii) the boundary-closeness weight w^{boundary} , which prevents confusion in vertices near human-object contact regions. Dropping either visibility coverage or mask quality would mean the quality score is wholly computed from the value of the other term, while the boundary-closeness term can be dropped by setting it to unity. The results in Tab. 8 indicate that dropping any of these components leads to worse PA-CD. This effectively highlights that all elements of our segmentation design are important for obtaining clean object point clouds from the combined LRM mesh and, consequently, high-quality HOI reconstructions.

4.4 Qualitative Evaluation on In-The-Wild Images

To assess MILO’s robustness, we further evaluate it qualitatively on in-the-wild images from PICO-db [5]. As illustrated in Fig. 5, MILO recovers human and

Segmentation Used	PA-CD _h (cm) ↓	PA-CD _o (cm) ↓	PA-CD _{h+o} (cm) ↓
P3-SAM [37]	7.30	55.86	14.68
Ours	6.85	20.74	9.36

Table 6: Comparison with P3-SAM. Results presented on InterCap [19].

Method Used	PA-CD _h (cm) ↓	PA-CD _o (cm) ↓	PA-CD _{h+o} (cm) ↓
Kabsch	7.16	19.98	7.53
Kabsch + ICP	6.96	18.97	7.45

Table 7: Template alignment ablation. Results presented on InterCap [19].

vis_	cover	mask_	quality	boundary_	dist	PA-CD _h (cm) ↓	PA-CD _o (cm) ↓	PA-CD _{h+o} (cm) ↓
✓		✓		✓		6.85	20.74	9.36
✗		✓		✓		7.62	22.81	9.82
✓		✗		✓		7.61	22.55	9.83
✓		✓		✗		7.57	21.75	9.77

Table 8: Ablation of point-cloud segmentation. Results are reported on InterCap [19]. The columns represent the visibility coverage (**vis_****cover**) and the mask quality (**mask_****quality**), together defining the viewpoint quality score Q , and the boundary-closeness weight (**boundary_****dist**). Removing any component degrades performance, confirming each term’s contribution to clean point cloud segmentation.

object meshes that are better aligned in 3D and exhibit more coherent contact patterns and relative positioning compared to prior work [5].

5 Conclusion

We introduced MILO, an approach for reconstructing human-object interactions from a single image. MILO leverages the visual capabilities of Large Reconstruction Models to predict a holistic human-object mesh. It then segments the mesh to human and object parts, fits the SMPL-H body to the human part and optionally the object template to the object part, if such a template is available. By treating the LRM output as a geometric scaffold, our method captures arrangement and proximity cues while remaining agnostic to object category and shape. Quantitative experiments on multiple benchmarks show that MILO surpasses existing HOI reconstruction methods despite using significantly less privileged information. Moreover, qualitative results on in-the-wild images demonstrate the generalization to diverse objects, poses, and interaction contexts. These results highlight the promise of LRMs for scalable, label-efficient HOI estimation.

Limitations and Future Work. One limitation of our approach is that the performance of MILO is bounded by the quality of the underlying LRM. However, given the quick progress in this space [57,59,61], we expect that the performance of LRMs will continue to improve. Based on our results, we also observe that our performance is hindered by the accuracy of the point cloud segmentation. We anticipate that with better methods for point-cloud segmentation, our approach will also benefit. Future work includes the refinement of LRM outputs especially for smaller objects, integrating stronger instance-level reasoning and segmentation to extract a better object mesh from the LRM output, and extending the framework to video and multi-person/multi-object interaction settings.

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